

- 17** A  $3.0\text{ W}$  point source emits sound waves at  $500\text{ Hz}$ . The amplitude of a sound wave with an intensity of  $0.1\text{ W m}^{-2}$  at  $500\text{ Hz}$  is  $3.0\text{ }\mu\text{m}$ .

What is the amplitude of the sound wave at a distance of  $5.0\text{ m}$  from the point source?

- A**  $0.29\text{ }\mu\text{m}$       **B**  $0.93\text{ }\mu$       **C**  $1.1\text{ }\mu\text{m}$       **D**  $1.9\text{ }\mu\text{m}$